

Database Management Systems

CPS 3740

Lecture 09

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Outline

- Exam Policy
- Quick Midterm1 Review

Exam policy

- **Exam Format:** The exam will be conducted **in person** in the classroom using the **LockDown Browser**.
- **Preparation:** Bring your laptop and charger. Ensure your laptop is updated and **test the LockDown Browser using the practice item before the exam.**
- **Punctuality:** **Arrive at least 5 minutes early.** Attendance will be taken before the exam begins.
- **Prohibited Items:** Headphones, mobile phones, and other electronic devices are not allowed during the exam. **Mobile phones may only be used for Canvas login verification** before the exam starts and must then be stored away.
- **Room Policy:** **Once the exam begins, students cannot leave the room and return to continue their exam.** Leaving the exam room will indicate that you have completed your exam.

Midterm 1 coverage

- Midterm 1 coverage
 - Cover day 1 – 8 content(exclude SQL injection)
 - SQL injection and day 9 content belong to the scope of the midterm 2 and final exam.
- Practice exercises in the **dreamhome_SQL_exercises_answers.pdf** file, located in the OneDrive **tools** folder.

Midterm 1 format

- 75 minutes
 - Focus on your own questions. Do NOT help others
 - You won't have time to run on the command line.
 - **No extension! The same as interview in the future. → Training**
- Multiple choices, TRUE/FALSE, ESSAY
- Ignore any typo.
- Give tables, pick the correct SQL query
- Give tables and SQL, PHP, pick the correct output
- SQL query and PHP will be around 65%
 - Some questions will take more than 90 seconds
- Conceptual questions will be around 35%
 - Spend less than 45 seconds on each conceptual question

Midterm 1 coverage - Concepts

- Concepts of the database management systems
- Client-server, 3-tier architecture
- DBMS functions
- Client-server – putty, FileZilla, ssh, mysql, http
- Project - HTML, CGI, PHP MySQL functions, cookie
- Roles related to database
- Client/server

Midterm 1 coverage - project

- Web database, 3-tier client/server
 - browser, web server, database server
 - Examples of PC, Web server, database server
 - Which machine or resources the program runs
- HTML inputs, PHP MySQL, cgi-bin
- Linux commands

Midterm 1 coverage - SQL

- SELECT, SELECT, DISTINCT, WHERE, =, LIKE
 - NULL values, operations with AND, OR
 - WHERE conditions (compound)
 - GROUP BY, HAVING
 - AGGREGATE functions – MIN, MAX, SUM, COUNT, AVG
 - Distinct, alias
 - Set operations
 - ORDER BY
 - Compound conditions
 - JOIN multiple tables
 - Subquery – find who has the maximum, minimum salary
- Create view
- Not including CREATE table, INSERT, UPDATE, DROP, SQL Injection, Outer join
- Lab exercise, homework, dreamhome exercises pdf file

Review Linux commands

- Review the Linux command paper
- Permission mode
- Commands: ls, ps, more, mkdir, chmod, cd, rmdir, rm, pwd, mv, cp, man, kill, grep
- File system
 - List files with all details
 - Create folder
 - . and ..
 - Change file/folder name
 - Move a file/folder
 - Delete a file/folder
 - Change file/folder permission
 - Show current full path
 - Change folder location

Midterm Topics -SQL

- PHP MySQL functions
- SQL
 - SQL
 - NULL and logic operations
 - WHERE, GROUP BY, ORDER BY, IN, LIKE, NOT
 - Aggregation
 - Subquery
 - JOINS
 - Compound conditions

Exercise #1

- Run the following statements at mysql on imc.kean.edu

> use dreamhome

> show tables;

1. List all the hotel names at Paris city

- select hotelname from Hotel where city='Paris';

2. List each hotel's total revenue (per night), if all their rooms were sold.

- select hotelname,sum(price) from Hotel,Room where Hotel.hotelno=Room.hotelno group by hotelname;

Exercise #2

1. List the total number of rooms that hotel “Latham” has
 - `select count(*) from Room, Hotel where Hotel.hotelno=Room.hotelno and Hotel.hotelname='Latham';`
2. List the total number of room for each hotel
 - `select hotelname,count(*) from Room, Hotel where Hotel.hotelno=Room.hotelno group by hotelname;`

Exercise #3

- List the hotel name that doesn't have family room type. (Answers: Phoenix Park, Watergate)
- Will this SQL work?
 - `select hotelname from Hotel,Room where Hotel.hotelno=Room.hotelno and type !='family';`
- Find the hotel has family room type
 - `select hotelno from Room where type ='family';`
- Find hotel name not in the previous list
 - `select hotelname from Hotel where hotelno not in (select hotelno from Room where type ='family');`

More Exercises

- SQL exercise posted on the Google drive
 - Exercise pdf file: 1-68, 74-77, 116
- SELECT: 17, 18, 19, 25
- Aggregate, GROUP BY: 26, 31, 40, 41, 42
- Subquery: 45, 48, 53, 57
- Joins: 58, 62, 67, 68
- NULL: 116, 119

Leetcode

- The easy and medium questions
- <https://leetcode.com/problemset/database/>

Conclusion

- Study for midterm 1
- Must do all the exercises, NOT just read the slides
- Arrive classroom 5 minutes earlier